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Inventor(s)	Wixey; Matthew et al.

Surgical instruments with switches for deactivating and/or identifying stapler cartridges

Abstract

Surgical stapling instruments include mechanisms for identifying and/or deactivating stapler cartridge for use with the instruments. The stapling instrument includes a drive member for actuating a staple cartridge and a locking member movable from a disabled position permitting distal translation of the drive member through a staple firing stroke, to a locking position inhibiting distal translation of the drive member through the staple firing stroke. The staple cartridge may include a switch for maintaining the locking member in the disabled position. The switch may be further configured to operate as a reload detection mechanism for determining the type of reload present in the surgical stapling instrument.

Inventors: Wixey; Matthew (San Jose, CA), Patel; Atal (Mission Viejo, CA), Jasemian; Babak D. (Trabuco Canyon, CA), Ragosta; Nicholas (San Francisco, CA)

Applicant: INTUITIVE SURGICAL OPERATIONS, INC. (Sunnyvale, CA)

Family ID: 71100560

Assignee: INTUITIVE SURGICAL OPERATIONS, INC. (Sunnyvale, CA)

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation of U.S. application Ser. No. 17/414,805, filed on Jun. 16, 2021, which is the National Stage of International Application No. PCT/US2019/066513 filed Dec. 16, 2019, which claims the benefit of U.S. Provisional Application No. 62/783,429, filed Dec. 21, 2018, the entire disclosure of each are incorporated herein by reference for all purposes.

BACKGROUND

(1) The field of the present disclosure relates to medical instruments, and more particularly to tissue sealing instruments for use in surgeries. Even more particularly, the present disclosure relates to a surgical stapling instrument having a novel switch-activated lockout mechanism to prevent firing of a surgical stapling instrument while a spent stapler cartridge remains in place on the jaw. The present disclosure further relates to a surgical stapling instrument configured for use with a surgical system having a control unit configured to identify the type and size of a reload installed in the surgical stapling instrument.

(2) Minimally invasive medical techniques are intended to reduce the amount of extraneous tissue that is damaged during diagnostic or surgical procedures, thereby reducing patient recovery time, discomfort, and deleterious side effects. One effect of minimally invasive surgery, for example, is reduced post-operative hospital recovery times. The average hospital stay for a standard open surgery is typically significantly longer than the average stay for an analogous minimally invasive surgery (MIS). Thus, increased use of MIS could save millions of dollars in hospital costs each year. While many of the surgeries performed each year in the United States could potentially be performed in a minimally invasive manner, only a portion of the current surgeries uses these advantageous techniques due to limitations in minimally invasive surgical instruments and the additional surgical training involved in mastering them.

(3) Improved surgical instruments such as tissue access, navigation, dissection and sealing instruments have enabled MIS to redefine the field of surgery. These instruments allow surgeries and diagnostic procedures to be performed with reduced trauma to the patient. A common form of minimally invasive surgery is endoscopy, and a common form of endoscopy is laparoscopy, which is minimally invasive inspection and surgery inside the abdominal cavity. In standard laparoscopic surgery, a patient's abdomen is insufflated with gas, and cannula sleeves are passed through small (approximately one-half inch or less) incisions to provide entry ports for laparoscopic instruments.

(4) Laparoscopic surgical instruments generally include an endoscope (e.g., laparoscope) for viewing the surgical field and tools for working at the surgical site. The working tools are typically similar to those used in conventional (open) surgery, except that the working end or end effector of each tool is separated from its handle by an extension tube (also known as, e.g., an instrument shaft or a main shaft). The end effector can include, for example, a clamp, grasper, scissor, stapler, cautery tool, linear cutter, or needle holder.

(5) To perform surgical procedures, the surgeon passes working tools through cannula sleeves to an internal surgical site and manipulates them from outside the abdomen. The surgeon views the procedure from a monitor that displays an image of the surgical site taken from the endoscope. Similar endoscopic techniques are employed in, for example, arthroscopy, retroperitoneoscopy, pelviscopy, nephroscopy, cystoscopy, cisternscopy, sinoscopy, hysteroscopy, urethroscopy, and the like.

(6) Minimally invasive telesurgical systems are being developed to increase a surgeon's dexterity when working on an internal surgical site, as well as to allow a surgeon to operate on a patient from a remote location (outside the sterile field). In a telesurgery system, the surgeon is often provided with an image of the surgical site at a control console. While viewing a three dimensional image of the surgical site on a suitable viewer or display, the surgeon performs the surgical procedures on the patient by manipulating master input or control devices of the control console, which in turn control motion of the servo-mechanically operated slave instruments.

(7) The servomechanism used for telesurgery will often accept input from two master controllers (one for each of the surgeon's hands) and may include two or more robotic arms. A surgical instrument is mounted on each of the robotic arms. Operative communication between master controllers and associated robotic arm and instrument assemblies is typically achieved through a control system. The control system typically includes at least one processor that relays input commands from the master controllers to the associated robotic arm and instrument assemblies and back in the case of, for example, force feedback or the like. One example of a robotic surgical system is the DA VINCI™ system commercialized by Intuitive Surgical, Inc. of Sunnyvale, California.

(8) A variety of structural arrangements have been used to support the surgical instrument at the surgical site during robotic surgery. The driven linkage or "slave" is often called a robotic surgical manipulator, and exemplary linkage arrangements for use as a robotic surgical manipulator during minimally invasive robotic surgery are described in U.S. Pat. Nos. 7,594,912, 6,758,843, 6,246,200, and 5,800,423, the full disclosures of which are incorporated herein by reference in their entirety for all purposes. These linkages often manipulate an instrument holder to which an instrument having a shaft is mounted. Such a manipulator structure can include a parallelogram linkage portion that generates motion of the instrument holder that is limited to rotation about a pitch axis that intersects a remote center of manipulation located along the length of the instrument shaft. Such a manipulator structure can also include a yaw joint that generates motion of the instrument holder that is limited to rotation about a yaw axis that is perpendicular to the pitch axis and that also intersects the remote center of manipulation. By aligning the remote center of manipulation with the incision point to the internal surgical site (for example, with a trocar or cannula at an abdominal wall during laparoscopic surgery), an end effector of the surgical instrument can be positioned safely by moving the proximal end of the shaft using the manipulator linkage without imposing potentially hazardous forces against the abdominal wall. Alternative manipulator structures are described, for example, in U.S. Pat. Nos. 6,702,805, 6,676,669, 5,855,583, 5,808,665, 5,445,166, and 5,184,601, the full disclosures of which are incorporated herein by reference in their entirety for all purposes.

(9) During the surgical procedure, the telesurgical system can provide mechanical actuation and control of a variety of surgical instruments or tools having end effectors that perform various functions for the surgeon, for example, holding or driving a needle, grasping a blood vessel, dissecting tissue, or the like, in response to manipulation of the master input devices. Manipulation and control of these end effectors is a particularly beneficial aspect of robotic surgical systems. For this reason, it is desirable to provide surgical tools that include mechanisms that provide two or three degrees of rotational movement of an end effector to mimic the natural action of a surgeon's wrist. Such mechanisms should be appropriately sized for use in a minimally invasive procedure and relatively simple in design to reduce possible points of failure. In addition, such mechanisms should provide an adequate range of motion to allow the end effector to be manipulated in a wide variety of positions.

(10) Surgical instruments are often deployed into restrictive body cavities (e.g., through a cannula to inside the pelvis). Accordingly, it is desirable for the surgical instrument to be both compact and maneuverable for best access to and visibility of the surgical site. Known surgical instruments, however, may fail to be both compact and maneuverable. For example, known surgical instruments

may lack maneuverability with respect to multiple degrees of freedom (e.g., roll, pitch, and yaw) and associated desired ranges of motion.

(11) Surgical clamping and cutting instruments (e.g., non-robotic linear clamping, stapling, and cutting devices, also known as surgical staplers; and electrosurgical vessel sealing devices) have been employed in many different surgical procedures. For example, a surgical stapler can be used to resect a cancerous or anomalous tissue from a gastro-intestinal tract. Many known surgical clamping and cutting devices, including known surgical staplers, have opposing jaws that clamp tissue and an articulated knife to cut the clamped tissue.

(12) Many surgical clamping and cutting instruments include an instrument shaft supporting an end effector to which a replaceable stapler cartridge is mounted. An actuation mechanism articulates the stapler cartridge to deploy staples from the stapler cartridge to staple tissue clamped between the stapler cartridge and an articulable jaw of the end effector. Different types of stapler cartridges can be used that have different staple lengths suitable for different tissues to be stapled.

(13) The use of replaceable stapler cartridges does, however, give rise to some additional issues. For example, prior to use, a suitable stapler cartridge having the correct staple length should be mounted to the end effector. If a stapler cartridge having an unsuitable staple length is mistakenly mounted to the end effector, the tissue may be stapled with the unsuitable length staples if the error is not detected and corrected prior to stapling of the tissue. As another example, if a previously used stapler cartridge is not replaced with a new stapler cartridge, the tissue clamped between the previously used stapler cartridge and the articulable jaw cannot be stapled due to the lack of staples to deploy. A similar problem can arise if no stapler cartridge is mounted to the end effector. The danger of firing a surgical stapling instrument while a spent stapler cartridge remains in place on the jaw has given rise to the development of various lockout mechanisms. However, incorporating lockout features typically increases the diameter of the end effector, increasing overall instrument size and making a given instrument less ideal for minimally invasive surgery.

(14) Accordingly, while the new telesurgical systems and devices have proven highly effective and advantageous, still further improvements would be desirable. In general, it would be desirable to have a relatively compact mechanism in place to prevent firing of a surgical stapling instrument while a spent stapler cartridge remains in place on the jaw. Additionally, it would be desirable to have a mechanism allowing a robotic surgical system to detect the type of stapler cartridge or reload that has been installed. Thus, a need exists for a reload detection mechanism that can detect: whether a stapler cartridge is mounted to the surgical instrument; whether the mounted stapler cartridge is unfired (i.e., fresh) or has already been fired; and the type of the mounted stapler cartridge mounted to the end effector to ensure that the mounted stapler cartridge has a suitable staple length for the tissue to be stapled.

SUMMARY

(15) The following presents a simplified summary of the claimed subject matter in order to provide a basic understanding of some aspects of the claimed subject matter. This summary is not an extensive overview of the claimed subject matter. It is intended to neither identify key or critical elements of the claimed subject matter nor delineate the scope of the claimed subject matter. Its sole purpose is to present some concepts of the claimed subject matter in a simplified form as a prelude to the more detailed description that is presented later.

(16) The present disclosure relates to surgical stapling instruments that have devices or mechanisms for identifying and/or deactivating disposable stapler cartridges for use with the stapling instruments. The stapling instrument includes a drive member for actuating a staple cartridge and a locking member movable from a disabled position permitting distal translation of the drive member through a staple firing stroke, to a locking position inhibiting distal translation of the drive member through the staple firing stroke. The staple cartridge may include a switch for maintaining the locking member in the disabled position. The switch may be further configured to operate as a reload detection mechanism for determining the type of reload present in the surgical stapling

instrument. In one embodiment, a surgical stapling instrument includes an end effector defining a longitudinal axis including a first jaw and a second jaw. The first jaw includes an anvil and, the second jaw is configured to receive a stapler cartridge having one or more staples. The surgical stapling instrument further includes a drive member configured to translate distally and an actuation mechanism configured to translate the drive member distally through the end effector. The surgical stapling instrument further includes a locking member movable from a disabled position permitting distal translation of the drive member to at least an axial position wherein the drive member engages at least one of the staples, to a locking position inhibiting distal translation of the drive member to said axial position. In the locking position, the locking member functions to deactivate the stapler cartridge by preventing firing of a surgical stapling instrument while a spent stapler cartridge remains in place on the jaw. This ensures that a surgeon will not attempt to clamp or seal tissue with a staple cartridge that has already been deployed and thus is unable to drive staples into the tissue.

(17) In embodiments, the locking member is maintained in the disabled position by a portion of the stapler cartridge. In embodiments, the portion of the stapler cartridge that maintains the locking member in the disabled position is a switch movably coupled to the stapler cartridge. In embodiments, the locking member moves in a first direction, and the switch is movable in a second direction different from the first direction. This is advantageous because it allows for maintenance of reduced instrument diameter, as the switch and the locking member will not be contained in the same space within the surgical instrument once actuation has occurred.

(18) In embodiments, the locking member includes a distal portion configured to contact the switch, and a distal drive member-engaging portion.

(19) In embodiments, the drive member includes one or more inclined distal surfaces or ramps. In embodiments, upon distal advancement of the drive member, the inclined distal surface(s) of the drive member engage the switch and moves the switch from a first position to a second position. In the first position, the switch maintains the locking member in the disabled position, permitting the drive member to translate distally through the end effector. As the inclined distal surfaces of the drive member contact the switch, they move the switch into the second position, wherein the switch no longer engages the locking member. The locking member, which is preferably biased towards the locking position, will then automatically move into the locking position.

(20) The switch is preferably retained in the second position once it has been moved into this position. Thus, the stapler cartridge can only be used once. As soon as the drive member actuates the staples and moves the switch into the second position, the locking member moves into the locking position and remains in this position so that the drive member can no longer translate distally to actuate the stapler cartridge.

(21) In embodiments, the switch includes a cutout of a predetermined height configured to be engaged by the inclined distal surface of the drive member. In embodiments, engagement of the inclined distal surface of the drive member with the cutout creates a detectable resistance readable by a control unit of a surgical system to detect a given reload size or type.

(22) In embodiments, the switch includes a stationary portion and a movable portion, the stationary portion configured to be separated from the movable portion by shearing along an axis upon contact by the drive member. In embodiments, the engagement between an inclined distal surface of the drive member with the switch creates a detectable resistance, the detectable resistance readable by a control unit of a surgical system to detect a given reload size or type.

(23) In embodiments, the locking member pivots between the disabled position and the locking position. In embodiments, the locking member pivots about a pivot point that is laterally offset from the longitudinal axis of the end effector. In embodiments, the locking member pivots in a direction substantially perpendicular to the longitudinal axis defined by the end effector.

(24) In embodiments, the drive member includes a first portion that translates through a channel in the first jaw. In embodiments, the actuation mechanism includes a coil that applies a distal force to

the first portion of the drive member.

(25) In embodiments, the surgical stapling instrument further includes an elongated shaft, the end effector mounted on a distal end portion of the elongated shaft.

(26) In embodiments, the surgical stapling instrument further includes an articulation mechanism configured to articulate the end effector relative to the elongate shaft. In embodiments, the surgical stapling instrument further includes an actuator operatively connected to the actuation mechanism. In embodiments, the actuator includes a movable handle of a handle assembly provided at a proximal end portion of the surgical instrument. In embodiments, the actuator includes a control device of a robotic surgical system. In embodiments, the drive member includes a knife configured to cut tissue grasped between the first and second jaw.

(27) In another aspect, the present disclosure relates to a surgical stapling instrument including an end effector defining a longitudinal axis including a first jaw and a second jaw, the first jaw including an anvil. The surgical stapling instrument further includes a stapler cartridge having one or more staples and a switch. The second jaw is configured to receive the stapler cartridge. The surgical stapling instrument further includes a drive member configured to translate distally and an actuation mechanism configured to translate the drive member distally through the end effector. The drive member is configured to contact the switch of the stapler cartridge at an axial position of the drive member relative to the end effector. The switch is configured to provide a detectable resistance upon engagement of the drive member at said axial position. This detectable resistance is advantageous because it may provide information for a reload detection mechanism that can detect: whether a stapler cartridge is mounted to the surgical instrument; whether the mounted stapler cartridge is unfired (or fresh) or has already been fired; and the type of the mounted stapler cartridge mounted to the end effector to ensure that the mounted stapler cartridge has a suitable staple length for the tissue to be stapled.

(28) In embodiments, the surgical stapling instrument further includes a lockout assembly including a locking member movable in a first direction from a disabled position permitting distal translation of the drive member through a staple firing stroke, to a locking position inhibiting distal translation of the drive member through the staple firing stroke. the surgical stapling instrument further includes a switch movable in a second direction different from the first direction, from a first position and second position, wherein when the switch is in the first position the switch maintains the locking member in the disabled position, and wherein when the switch is in the second position the switch disengages from the locking member.

(29) In embodiments, the drive member includes one or more inclined distal ramps and the switch has a contact portion configured to contact the one or more distal ramps upon distal translation of the drive member through the end effector. In embodiments, the contact portion of the switch is disposed at a predetermined height such that the inclined distal ramp of the drive member is located at the axial position upon contact with the contact portion.

(30) In certain embodiments, the surgical instrument includes two or more staple cartridges. Each of the staple cartridges includes a switch having a contact portion configured to contact the one or more distal ramps of the drive member. Each of the contact portions of the switches is located at a different height relative to the end effector (and the drive member). Since the distal ramp(s) of the drive member are inclined, the drive member will contact each of the switches of the different staple cartridges at different axial positions of the drive member relative to the staple cartridge. This contact is detectable by a control unit or other suitable mechanism such that each of the staple cartridges may be identified by the control unit.

(31) In embodiments, the surgical instrument is operatively coupled to a control unit, the control unit configured to process the detectable resistance to identify a type of reload present in the surgical stapling instrument.

(32) In certain embodiments, the switch is configured to provide the detectable resistance to the control until upon engagement of the drive member with the contact portion of the switch. I

(33) In other embodiments, the switch includes a detachable portion configured to detach from the remainder of the switch upon contact with the drive member. In these embodiments, the control unit detects resistance upon detachment of the detachable portion, thereby identifying the stapler cartridge. In an exemplary embodiment, the switch includes a stationary portion and a movable portion, the stationary portion being configured to be separated from the movable portion by shearing along an axis upon contact by the drive member.

(34) In another aspect, the present disclosure relates to a surgical stapling instrument including an end effector defining a longitudinal axis including a first jaw and a second jaw. The first jaw includes an anvil and, the second jaw is configured to receive a stapler cartridge having one or more staples. The surgical stapling instrument further includes a drive member configured to translate distally and an actuation mechanism configured to translate the drive member distally through the end effector. The surgical stapling instrument further includes a lockout assembly including a locking member movable in a first direction from a disabled position permitting distal translation of the drive member through a staple firing stroke, to a locking position inhibiting distal translation of the drive member through the staple firing stroke. The drive member is configured to contact a switch at an axial position of the drive member relative to the end effector, and wherein the switch is configured to provide a detectable resistance upon engagement of the drive member at said axial position.

(35) In embodiments, the locking member is maintained in the disabled position by a portion of the stapler cartridge. In embodiments, the portion of the stapler cartridge that maintains the locking member in the disabled position comprises the switch. In embodiments, the locking member moves in a first direction, and the switch is movable in a second direction different from the first direction.

(36) In embodiments, the drive member includes one or more inclined distal ramps and the switch has a contact portion configured to contact the one or more distal ramps upon distal translation of the drive member through the end effector. In embodiments, the contact portion of the switch is disposed at a predetermined height such that the inclined distal ramp of the drive member is located at the axial position upon contact with the contact portion.

(37) In embodiments, the surgical instrument is operatively coupled to a surgical system including a control unit, the control unit configured to process the detectable resistance to identify a type of reload present in the surgical stapling instrument.

(38) In embodiments, the switch includes a stationary portion and a movable portion. The stationary portion is configured to be separated from the movable portion by shearing along an axis upon contact by the drive member.

(39) In embodiments, the surgical instrument is operatively coupled to a surgical system including a control unit. The control unit is configured to process the detectable resistance to identify a type of reload present in the surgical stapling instrument.

(40) In yet another aspect, the present disclosure relates to a surgical stapling instrument including an end effector defining a longitudinal axis including a first jaw and a second jaw. The first jaw includes an anvil and, the second jaw is configured to receive a stapler cartridge having one or more staples. The surgical stapling instrument further includes a drive member configured to translate distally and an actuation mechanism configured to translate the drive member distally through the end effector. The surgical stapling instrument further includes a lockout assembly including a locking member movable in a first direction from a disabled position permitting distal translation of the drive member through a staple firing stroke, to a locking position inhibiting distal translation of the drive member through the staple firing stroke. The drive member is configured to contact a first switch at an axial position of the drive member relative to the end effector, and a second switch, and wherein the first switch is configured to provide a detectable resistance upon engagement of the drive member at said axial position. This is advantageous because the detectable resistance provided upon engagement of the drive member with the first switch for reload detection may occur at a more proximal position within the surgical instrument, such as a proximal tail

portion of the cartridge.

(41) In embodiments, the locking member is maintained in the disabled position by a portion of the stapler cartridge. In embodiments, the portion of the stapler cartridge that maintains the locking member in the disabled position comprises the second switch. In embodiments, the locking member moves in a first direction, and the first switch and second switch are both movable in a second direction different from the first direction.

(42) In embodiments, the drive member includes one or more inclined distal ramps and the first switch and second switch have contact portions configured to contact the one or more distal ramps upon distal translation of the drive member through the end effector. In embodiments, the contact portion of the first switch is disposed at a predetermined height such that the inclined distal ramp of the drive member is located at the axial position upon contact with the contact portion.

(43) In embodiments, the first switch is formed on a proximal tail portion of the stapler cartridge.

(44) In embodiments, the surgical instrument is operatively coupled to a surgical system including a control unit. The control unit is configured to process the detectable resistance to identify a type of reload present in the surgical stapling instrument.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The above and other aspects, features, and advantages of the present surgical instruments having a locking mechanism will become more apparent in light of the following detailed description when taken in conjunction with the accompanying drawings in which:

(2) FIG. 1 is a perspective view of an illustrative surgical instrument having an end effector mounted to an elongated shaft, and an actuation mechanism;

(3) FIG. 2 is a perspective view of the distal end portion of an illustrative surgical instrument in accordance with the present disclosure with the jaws in the open position;

(4) FIG. 3 is an exploded view of the end effector of FIG. 2;

(5) FIG. 4 is an exploded view of the cartridge, surgical fasteners, staple drivers, and switches which form part of the cartridge assembly of FIG. 3;

(6) FIG. 5 depicts a partial cross-sectional side view of the end effector of a surgical stapling instrument including a lockout assembly in accordance with an embodiment of the present disclosure having an unfired stapler cartridge installed;

(7) FIG. 6 depicts a top view of a locking member in accordance with the embodiment of FIG. 5 in the unlocked position;

(8) FIG. 6A depicts a top view of a locking member in accordance with the embodiment of FIG. 5 in the locked position;

(9) FIG. 7 depicts a perspective view of the proximal end of a stapler cartridge in accordance with the embodiment of FIG. 5 having a lockout assembly including a switch for enabling and disabling the locking member;

(10) FIG. 7A depicts a perspective view of a switch in accordance with the embodiment of FIG. 5;

(11) FIG. 8 is a partial perspective view of the end effector including a lockout assembly in accordance with the embodiment of FIG. 5 showing a wedge of the drive member contacting a switch;

(12) FIG. 9 is a partial perspective view of an end effector in accordance with the embodiment of FIG. 5 showing a drive member moving distally towards a switch;

(13) FIG. 10 is a partial perspective view of an end effector in accordance with the embodiment of FIG. 5 showing a drive member contacting a switch upon translating distally;

(14) FIG. 11 is a partial perspective view of an end effector in accordance with the embodiment of FIG. 5 showing a wedge of the drive member engaging and pushing a switch upwards during

actuation;

(15) FIG. 12 is a partial perspective view of an end effector in accordance with the embodiment of FIG. 5 showing a switch pushed completely upwards into a raised position, and the drive member translating distally underneath the switch;

(16) FIG. 13 is a top view of an end effector in accordance with the embodiment of FIG. 5 showing a drive member translated distally, a switch in the raised position, and a locking member that is enabled;

(17) FIGS. 13A-13D are perspective views of the proximal ends of a series of stapler cartridges, each stapler cartridge containing a unique switch for reload detection;

(18) FIG. 14 is a cross-sectional side of a two-part clevis of the surgical instrument of FIG. 1;

(19) FIG. 15 is a perspective view of the end portion of an illustrative surgical instrument with parts removed;

(20) FIG. 16 is a perspective view of a drive member in accordance with the illustrative surgical instrument of FIG. 1;

(21) FIG. 17A is a cross-sectional perspective view of the actuation mechanism for a drive member in accordance with the surgical instrument of FIG. 1;

(22) FIG. 17B is a cross-sectional side view of the actuation mechanism for a drive member in accordance with the surgical instrument of FIG. 1;

(23) FIG. 18A shows a movable lower jaw of an illustrative surgical instrument in an open configuration;

(24) FIG. 18B shows a movable lower jaw of an illustrative surgical instrument pivoting towards a closed position;

(25) FIG. 18C shows a movable lower jaw of an illustrative surgical instrument in a closed position;

(26) FIG. 19 is a perspective view of a switch usable with an alternative embodiment of the present disclosure;

(27) FIG. 20 is a perspective view of a stapler cartridge including the switch of FIG. 19 in a first position before actuation;

(28) FIG. 21 is a perspective view of a stapler cartridge including the switch of FIG. 19 after actuation;

(29) FIG. 22 is a partial side view of the proximal portion of the stapler cartridge of FIG. 21;

(30) FIG. 23 is a cross-sectional top view of an illustrative end effector with a switch and stapler cartridge in accordance with FIG. 20 installed before actuation;

(31) FIG. 24 is a partial side view of a portion of the illustrative end effector of FIG. 23 showing a shuttle contacting the switch upon actuation of the surgical instrument;

(32) FIG. 25 is a partial side view with parts removed of a portion of the illustrative end effector of FIG. 23 showing the switch after it has been engaged by a shuttle upon actuation;

(33) FIG. 26 shows an alternative embodiment having an illustrative stapler cartridge containing a first switch for reload detection, and a second switch for engaging locking member;

(34) FIG. 27 illustrates a top view of an operating room employing a robotic surgical system utilizing aspects of the present disclosure; and

(35) FIG. 28 illustrates a simplified side view of a robotic arm assembly that is usable with various aspects of the present disclosure.

DETAILED DESCRIPTION

(36) Particular embodiments of the present surgical instruments are described hereinbelow with reference to the accompanying drawings; however, it is to be understood that the disclosed embodiments are merely exemplary of the disclosure and may be embodied in various forms. Therefore, specific structural and functional details disclosed herein are not to be interpreted as limiting, but merely as a basis for the claims and as a representative basis for teaching one skilled in the art to variously employ the present disclosure in virtually any appropriately detailed

structure. Well-known functions or constructions are not described in detail to avoid obscuring the present disclosure in any unnecessary detail.

(37) While the following disclosure is presented with respect to a linear surgical stapler where staples are sequentially fired, it should be understood that the features of the presently described surgical instruments may be readily adapted for use in any type of surgical clamping, cutting, or sealing instruments. The surgical clamping and cutting instrument may be a minimally invasive (e.g., laparoscopic) instrument or an instrument used for open surgery.

(38) Additionally, the features of the presently described surgical stapling instruments may be readily adapted for use in surgical instruments that are activated using any technique within the purview of those skilled in the art, such as, for example, manually activated surgical instruments, powered surgical instruments (e.g., electro-mechanically powered instruments), robotic surgical instruments, and the like.

(39) FIG. 1 is a perspective view of an illustrative surgical instrument **100** in accordance with embodiments of the present disclosure having a handle assembly **102**, and an end effector **110** mounted on an elongated shaft **106**. End effector **110** includes a stationary jaw **111** and a moveable jaw **112**. Handle assembly **102** includes a stationary handle **102a** and a moveable handle **102b** which serves as an actuator for surgical instrument **100**.

(40) In certain embodiments, handle assembly **102** may include input couplers (not shown) instead of, or in addition to, the stationary and movable handles. The input couplers provide a mechanical coupling between the drive tendons or cables of the instrument and motorized axes of the mechanical interface of a drive system. The input couplers may interface with, and be driven by, corresponding output couplers (not shown) of a telesurgical surgery system, such as the system disclosed in U.S. Pub. No. 2014/0183244A1, the entire disclosure of which is incorporated by reference herein. The input couplers are drivingly coupled with one or more input members (not shown) that are disposed within the instrument shaft **106**. The input members are drivingly coupled with the end effector **110**. Suitable input couplers can be adapted to mate with various types of motor packs (not shown), such as the stapler-specific motor packs disclosed in U.S. Pat. No. 8,912,746, or the universal motor packs disclosed in U.S. Pat. No. 8,529,582, the disclosures of both of which are incorporated by reference herein in their entirety. Further details of known input couplers and surgical systems are described, for example, in U.S. Pat. Nos. 8,597,280, 7,048,745, and 10,016,244. Each of these patents is hereby incorporated by reference in its entirety.

(41) Actuation mechanisms of surgical instrument **100** may employ drive cables that are used in conjunction with a system of motors and pulleys. Powered surgical systems, including robotic surgical systems that utilize drive cables connected to a system of motors and pulleys for various functions including opening and closing of jaws, as well as for movement and actuation of end effectors are well known. Further details of known drive cable surgical systems are described, for example, in U.S. Pat. Nos. 7,666,191 and 9,050,119 both of which are hereby incorporated by reference in their entireties. While described herein with respect to an instrument configured for use with a robotic surgical system, it should be understood that the wrist assemblies described herein may be incorporated into manually actuated instruments, electro-mechanical powered instruments, or instruments actuated in any other way.

(42) FIG. 2 shows the distal end portion of surgical instrument **100**, including an end effector **110** defining a longitudinal axis X-X and having a first jaw **111**, a second jaw **112**, a clevis **140** for mounting jaws **111**, **112** to the instrument). In certain embodiments, second jaw **112** is a movable jaw configured to move from an open position to a closed position relative to first jaw **111**. In other embodiments, first jaw **111** is a movable jaw configured to move between open and closed positions relative to second jaw **112**. In still other embodiments, both jaws **111**, **112** are movable relative to each other.

(43) First jaw **111** includes an anvil **115** having staple-forming pockets **116**. Second jaw **112** is configured to move from an open position to a closed position relative to stationary jaw **111**. In the

open position, a fresh stapler cartridge **122** such as the exemplary one shown in FIG. **3** (sometimes referred to as a reload) can be loaded into movable jaw **112** and tissue may be positioned between the jaws **111**, **112**. In the closed position, jaws **111**, **112** cooperate to clamp tissue such that cartridge **122** and the anvil **115** are in close cooperative alignment.

(44) As shown in FIG. **4**, stapler cartridge **122** may include a plurality of staples **124** supported on corresponding staple drivers **126** provided within respective staple retention openings **127** formed in stapler cartridge **122**. In embodiments, a shuttle **123** (see FIG. **3**) having an inclined distal portion **125** sequentially acts on staple drivers **126** upon distal movement of the drive member **150**, camming staple drivers **126** upwardly, thereby moving staples **124** into deforming contact with anvil **115**. In embodiments, shuttle **123** may be included within stapler cartridge **122**. In embodiments, inclined distal portions **125** may be integrated with drive member **150** as seen in FIG. **3**. In embodiments, stapler cartridge **122** further includes one or more switches **191** positioned on the proximal side of the proximal-most pusher **126p** within stapler cartridge **122**. The functionality of switches **191** will be described in more detail below.

(45) As shown in FIGS. **3** and **4**, end effector **110** may also include a lockout assembly including locking member **170**, switch **191**, and spring **178**. Locking member **170** includes a distal, switch-contacting portion **172** and a proximal engagement portion **174** (See FIG. **5**). End effector **110** may also include a drive member **150** configured to translate distally and retract proximally through the end effector and includes an integrated shuttle **123** having an inclined distal portion **125** formed thereon. As seen in FIG. **2**, upper shoe **152** of drive member **150** is substantially aligned with and translates through a channel **118** in fixed jaw **111**, while lower shoe **154** (see FIG. **16**) of drive member **150** is substantially aligned with and translates through a channel **119** and below jaw **112**. The details of the drive member and actuation will be described below.

(46) FIG. **5** shows a portion of an illustrative surgical instrument with an unfired reload installed, including stapler cartridge **122**, spring **178**, locking member **170**, and switch **191**. When an unfired stapler cartridge is installed, as shown in FIG. **5**, switch **191** is in a first unraised position. In a fresh, unfired stapler cartridge, switch **191** is in contact with distal portion **172** of locking member **170**, keeping distal portion **172** held within channel **119** and proximal engagement portion **174** outside of channel **119**. When locking member **170** is in this disabled position, distal translation of drive member **150** is permitted, as locking member **170** will not obstruct movement of drive member **150** because engagement portion **174** is held out of alignment with channel **119**.

(47) FIGS. **6** and **6A** show a top view of locking member **170** in the unlocked or disabled position and the locked position, respectively.

(48) Locking member **170** pivots about a pivot point **179**, that is laterally offset from channel **119**, and is configured to move in a lateral direction, preferably substantially perpendicular to, the longitudinal axis of the end effector. Spring **178** biases engagement portion **174** of locking member **170** into channel **119** to lock the instrument. In the unlocked position of FIGS. **5** and **6**, switch **191** engages distal portion **172** of locking member **170**, overcoming the bias of spring **178** and holding engagement portion **174** out of channel **119**, permitting distal movement of drive member **150**. When switch **191** is no longer in contact with distal portion **172** of locking member **170**, spring **178** forces engagement portion **174** of locking member into channel **119**, where it obstructs distal movement of drive member **150**, as best seen in FIG. **6A**.

(49) FIGS. **7** and **7A** show an unfired stapler cartridge with switch **191** in the initial, pre-firing position. Switch **191** may be contained in stapler cartridge **122**, and may be substantially aligned with staple drivers **126** on a desired side of the stapler cartridge where it may engage distal portion **172** of locking member **170** (see FIG. **8**). Switch **191** includes a cutout **196** (see FIG. **13A**) configured to be engaged by inclined distal portion **125** of shuttle **123** upon distal advancement of drive member **150** when the surgical instrument **100** is actuated. In an unfired stapler cartridge, as shown in FIG. **7**, switch **191** is in an initial pre-firing position, where it rests on the bottom of a switch channel **129** within stapler cartridge **122**. Switch channel **129** extends upwardly towards

anvil **115**. As shown in FIG. 7A, switch **191** further includes detent **193**. Detent **193** is configured to provide mechanical resistance that must be overcome by drive member **150** in order to slide switch **191** from the initial position toward anvil **115**. This ensures that the lockout will not unintentionally activate as may happen if switch **191** freely slides in channel **129** (e.g., in the absence of detent **193**).

(50) FIG. 8 depicts inclined distal portions **125** of drive member **150** interfacing with switch **191** to disable locking member **170**. Distal portion **172** of locking member **170** is shown protruding into channel **119**, as switch **191** is blocking it from being laterally displaced.

(51) FIGS. 9-12 depict sequential cross-sectional views (with locking member **170** not shown) of a portion of an illustrative surgical instrument throughout distal translation of drive member **150** during actuation. Inclined distal portions **125** of drive member **150** are shown interfacing with switch **191** to move it from the unraised position, to the raised position in which locking member **170** becomes enabled.

(52) In FIG. 9, drive member **150** begins to translate distally along the longitudinal axis of the end effector and has not yet contacted switch **191**. In FIG. 10, inclined distal portion **125** of drive member **150** contacts switch **191**. As seen in FIG. 11, inclined distal portion **125** of drive member **150** then begins to force switch **191** upwards within switch channel **129** of stapler cartridge **122** as drive member **150** continues to translate distally. Switch **191** travels in a direction substantially perpendicular to the longitudinal axis of the end effector. In embodiments, switch **191** travels vertically, while locking member **170** travels laterally, allowing for maintenance of a reduced instrument diameter, as they will not compete for the same space within the instrument after actuation has occurred.

(53) As seen in FIG. 12, drive member **150** continues to travel distally and forces switch **191** into a fully raised position within channel **129**, allowing the entirety of drive member **150** to pass by switch **191** such that it may complete the firing stroke. Switch **191** is fitted into channel **129** such that it may not return to the unraised position once the drive member has traveled distally. As drive member **150** displaces switch **191**, distal portion **172** of locking member **170** is prevented from lateral movement by contact with drive member **150**.

(54) FIG. 13 depicts a top view of an illustrative surgical instrument during actuation. In FIG. 13, drive member **150** has translated past switch **191** and moved it into the fully raised position, enabling locking member **170**. In this configuration, drive member **150** may continue to travel distally to drive staples into the tissue grasped between jaws **111**, **112** and cut the stapled tissue. Upon retraction, a proximal surface on the proximal end of drive member **150** engages a proximal ramped surface **176** on locking member **170**, allowing drive member **150** to return to a position proximal of locking member **170**. However, once drive member **150** is positioned proximally of locking member **170**, if another attempt is made to actuate the instrument, drive member **150** will be obstructed by engagement portion **174** of locking member **150**, preventing actuation of an unloaded instrument.

(55) In another aspect of the present disclosure, the particular type of stapler cartridge (or reload) installed in the end effector may be detected based on the configuration of the switches **191**. Specifically, different stapler cartridges **122** may have switches **191** including a cutout **196** of a given height depending on the type of reload present within stapler cartridge **122**. FIGS. 13A-13D depict four different stapler cartridges **122** having switches **191** each including a cutout **196** of a given height depicted by arrows A-D respectively. As the length of cutout **196** increases, the distance that shuttle **123** must travel, as described above, before engaging switch **191** increases. Thus, as the length of cutout **196** increases, the contact point between inclined distal portion **125** and switch **191** moves upwards and towards the proximal end of inclined distal portion **125**. When the drive member **150** engages switch **191**, it creates a detectable resistance and urges switch **191** upwards towards a raised position as drive member **150** travels distally.

(56) In embodiments, a control unit of a robotic surgical system may be configured to detect the

axial position along a firing stroke at which the shuttle **123** engages a given switch **191** via detection of a detectable resistance, such as a torque spike, allowing the system to determine the type of reload presently installed. Varying the length of the cutout formed on a given switch will adjust the axial position at which the drive member engages the switch to create the detectable resistance. Based on the detected resistance, a control unit, operatively coupled with the actuation mechanism, determines the correct amount of forces to apply to the drive member depending upon the features of the detected stapler cartridge, including but not limited to, the number of staples contained therein, the size of the staples contained therein, and the geometry of the staples contained therein. An exemplary surgical stapler including a control unit of a surgical system that is operatively coupled to the actuation mechanism is described for example in International Application No. PCT/JS2017050747, the disclosure of which is hereby incorporated by reference in its entirety.

(57) Jaws **111**, **112** are attached to surgical instrument **100** via clevis **140**. See, FIG. **14**. Clevis **140** includes a proximal surface **140a** and a distal surface **140b**. Clevis **140** further includes upper clevis portion **142** and lower clevis portion **141** that cooperate when assembled to form protrusion **145** (best seen in FIG. **18A**) configured to engage tabs **113** (best seen in FIG. **18A** of jaw **111** to securely mount jaw **111** in a fixed position on instrument **100**. As seen in FIG. **14**, lower clevis portion **141** includes a pair of distally extending arms **147** for supporting movable jaw **112**. Arms **147** include opening **149** for receiving a pivot pin **130** defining a pivot axis around which jaw **112** pivots as described in more detail below. Lower clevis portion **141** also includes ramped groove **144** configured to guide a portion of an actuation coil **120** emerging from wrist **160**. Upper clevis portion **142** includes a complementary shaped ramped groove **146** that cooperates with ramped groove **144** of lower clevis portion **141** to form an enclosed channel **180** that guides coil **120** as it jogs upwards from wrist **160** towards distal surface **157** of upper shoe **152** of drive member **150**. In embodiments, channel **180** may include a first end **181** at a central portion of proximal surface **140a** and a second end **182** at a peripheral portion of distal surface **140b**. In embodiments, enclosed channel **180** may be substantially “S” shaped. Although shown as a two-part clevis, it should be understood that the clevis may be a unitary structure formed, for example, by molding, machining, 3-D printing, or the like.

(58) End effector **110** may be articulated in multiple directions by an articulation mechanism. In embodiments, the articulation mechanism may be a wrist **160** as shown, although other articulation mechanisms are contemplated. As seen in FIG. **15**, wrist **160** includes a plurality of articulation joints **162**, **164**, **166**, etc. that define a bore **167** through which an actuation mechanism (in embodiments, coil **120** and drive cable **171**, see FIG. **17A**) may pass. Upon exiting articulation wrist **160**, coil **120** enters and passes through channel **180** of clevis **140** (see FIG. **14**), ultimately engaging proximal surface **153** of upper shoe **152** of drive member **150**. Other articulation mechanisms within the purview of those skilled in the art may substitute for wrist **160**. One suitable articulation mechanism is described for example in U.S. Publication No. 2015/0250530, the disclosure of which is hereby incorporated by reference in its entirety.

(59) Upon actuation of the surgical instrument, drive member **150** is advanced distally through end effector **110** to move jaws **111**, **112** from the open position to the closed position, after which shuttle **123** and knife **128** are advanced distally through cartridge **122** to staple and cut tissue grasped between jaws **111**, **112**. Drive member **150** may be any structure capable of pushing at least one of a shuttle or a knife of a surgical stapling instrument with the necessary force to effectively sever or staple human tissue. Drive member **150** may be an I-beam, an E-beam, or any other type of drive member capable of performing similar functions. Drive member **150** is movably supported on the surgical stapling instrument **100** such that it may pass distally through cartridge **122** and upper fixed jaw **111** and lower jaw **112** when the surgical stapling instrument is fired (e.g., actuated).

(60) As seen in FIG. **16**, drive member **150** may include a body **151**, upper shoe **152**, lower shoe

154, and central portion **156**. Upper shoe **152** of drive member **150** is substantially aligned with and translates through a channel **118** in fixed jaw **111**, while lower shoe **154** of drive member **150** is substantially aligned with and translates through a channel **119** and below jaw **112**. Bore **158** is formed through upper shoe **152** to receive drive cable **171** as will be described in more detail below. Proximal surface **153** of upper shoe **152** is configured to be engaged by a coil **120** of actuation assembly **190** such that coil **120** may apply force to upper shoe **152** to advance drive member **150** distally, i.e., in the direction of arrow “A” in FIG. **17B**. A knife **128** may be formed on drive member **150** along the distal edge between upper shoe **152** and central portion **156**. In embodiments, inclined distal portions **125** may be formed on either side of drive member **150**.

(61) Actuation assembly **190** includes a drive cable **171**, a coil **120**, a sheath **121** surrounding coil **120**, and a drive rod **175**. Drive cable **171** includes an enlarged distal end **173**.

(62) As seen in FIGS. **17A** and **17B**, upper shoe **152** of drive member **150** includes a bore **158** into which drive cable **171** is routed. When assembling illustrative surgical instrument **100**, coil **120** and a protective sheath **121** are slipped over the free end of drive cable **171**. The free end of drive cable **171** is attached to a drive rod **175** securing coil **120** and the protective sheath **121** between drive member **150** and drive rod **175** as seen in FIG. **17B**. Sheath **121** may function to promote stability, smooth movement, and prevent buckling upon actuation of surgical instrument **100**. Sheath **121** may be made from polyimide, or any other suitable material having the requisite strength requirements such as various reinforced plastics, a nickel titanium alloy such as NITINOL™, poly para-phenyleneterphtalamide materials such as KEVLAR™ commercially available from DuPont. Other suitable materials may be envisioned by those of skill in the art. Enlarged distal end **173** of drive cable **171** resides within an enlarged distal portion **159** of bore **158** in upper shoe **152** of body **150**, such that the proximal face **157** of enlarged distal end **173** may apply a retraction force on upper shoe **152** when the drive cable **171** is pulled proximally, i.e., in the direction of arrow “B” in FIG. **17B**. Drive rod **175** is operationally connected to an actuator (e.g., movable handle **102b**), which allows distal translation and proximal retraction of actuation assembly **190**.

(63) In certain embodiments, the surgical instrument may be designed such that the drive member **150** is not retracted in the proximal direction after the staples have been fired. Those skilled in the art will recognize that in a manually actuated instrument, the actuator may be a movable handle, such as moveable handle **102b** shown in FIG. **1**; in a powered instrument the actuator may be a button (not shown) that causes a motor to act on the drive rod; and in a robotic system, the actuator may be a control device such as the control devices described below in connection with FIG. **28**. Any suitable backend actuation mechanism for driving the components of the surgical stapling instrument may be used. For additional details relating to exemplary actuation mechanisms using push/pull drive cables see, e.g., commonly owned International Application WO 2018/049217, the disclosure of which is hereby incorporated by reference in its entirety.

(64) During actuation of illustrative surgical instrument **100**, drive rod **175** applies force to coil **120**, thereby causing coil **120** to apply force to upper shoe **152** of drive member **150**, translating it distally (i.e., in the direction of arrow “A” in FIG. **17B**) initially closing jaws **111, 112** and then ejecting staples **124** from stapler cartridge **122** to staple tissue. After stapling is complete, drive rod **175** applies a force in the proximal direction to effect retraction of drive member. During retraction, enlarged distal end **173** of drive cable **171** is obstructed by wall **157** of enlarged portion **159** of bore **158**, causing drive cable **171** to apply force to upper shoe **152** of drive member **150**, thereby translating drive member **150** in the proximal direction. One of ordinary skill in the art will appreciate that drive member **150**, drive cable **171**, and drive rod **175** all move in unison and remain in the same relative position to each other.

(65) In the preferred embodiment, drive cable **171** advances drive member **150** through fixed jaw **111** (instead of through the staple cartridge jaw as in conventional surgical stapling instruments). Eliminating the internal channel for the actuation mechanism from the staple cartridge provides more space in the cartridge for the staples and for the reinforcing wall discussed above. In

alternative embodiments, coil **120** of actuation assembly **190** may be coupled with lower shoe **154** instead of upper shoe **152**. In these embodiments, coil **120** applies force to lower shoe **153** to advance drive member **150** distally through a channel (not shown) in the lower jaw **112**. In these embodiments, coil **120** will advance at least through a portion of lower jaw **112** and staple cartridge **122**.

(66) FIGS. **18A-C** depict fixed jaw **111** and movable jaw **112** of illustrative surgical instrument **100** sequentially moving from an open configuration to a closed configuration. As shown in FIG. **18A**, in the open configuration, drive member **150** is positioned proximally of cam surface **114** formed on movable jaw **112**. As drive member **150** translates in the distal direction “A” movable jaw **112** will rotate towards the closed position around pivot **117**.

(67) In FIG. **18B**, drive member **150** has come into contact with cam surface **114** of movable jaw **112**. As lower portion **154** of drive member **150** rides underneath cam surface **114**, drive member **150** pushes movable jaw **112**, causing it to pivot towards the closed position.

(68) FIG. **18C** illustrates jaws **111**, **112** in the closed position. Drive member **150** has translated distally past cam surface **114**. In this position, tissue is clamped, and further advancement of the drive member will sever and staple tissue.

(69) In an alternative embodiment shown in FIG. **19**, switch **191** is replaced with a switch **292** having a detachable portion configured for detachment from the remainder of switch **191** as drive member engages the switch **292**. In an exemplary embodiment, switch **292** is configured to be sheared into two separate pieces along a shear plane X'-X'. In such embodiments, the surgical system may detect the force applied by a shuttle in shearing switch **292** to determine the type of reload within a given stapler cartridge. Switch **292** may be any desired shape, and may be sheared at various angles or along any line substantially parallel to the path of the drive member as it translates through a firing stroke.

(70) In embodiments, the axial position at which switch **292** is contacted by a drive member may be adjusted by including a cutout of a predetermined height in a similar manner as above in connection with previously described embodiments to create a detectable resistance at a unique axial position. The detectable resistance may similarly be used for reload detection in a similar manner as described above. In embodiments in which switch **292** is shearable, it is envisioned that the design of switch **292** may be adjusted to shear under a specific amount of force from the drive member that also provides for a suitable amount of detectable resistance depending on the sensitivity of the control unit of the surgical system being used. Thus, the act of shearing switch **292** alone provides for a suitable detectable resistance that may be used by a control system to determine the type of stapler cartridge present by determining the axial position at which the drive member experiences the resistance associated with shearing switch **292**.

(71) FIG. **20** depicts the proximal portion of an illustrative stapler cartridge having two switches **292** in an unraised position. In this configuration, actuation has not yet occurred and switch **292** has not been engaged by an inclined distal portion of a shuttle or drive member having a shuttle integrated thereon. FIGS. **21** and **22** show switch **292** after it has been engaged by a shuttle upon actuation of the surgical instrument. In this position, switch **292** has been sheared, and a movable portion **293** of switch has been forced into a raised position by a shuttle, while a stationary portion **294** remains in a substantially the same position as when switch **292** was in the unraised position. As shown in FIG. **22**, stationary portion **294** of switch **292** may include a protrusion **295** configured to fit within a cutout **227** formed within a sidewall of stapler cartridge **222**, helping stationary portion **294** to remain substantially unmoved throughout actuation.

(72) FIGS. **23-25** sequentially depict actuation of an illustrative surgical instrument having an end effector with a stapler cartridge in accordance with FIG. **20** installed.

(73) In FIG. **23**, a fresh stapler cartridge **222** having switches **292** has been installed into end effector **210** of an illustrative surgical instrument. A locking member **270** is biased by a spring **278** towards a channel **219** through which drive member **250** is configured to pass. As in the

embodiments previously described, switch **292** in the unraised position maintains locking member **270** out of engagement with channel **219**, thereby allowing drive member **250** to translate distally upon actuation to cut tissue and drive staples and cut tissue.

(74) In FIG. **24**, an inclined distal portion **225** of a shuttle **223** is shown driving distally to engage switch **292**. Shuttle **223** may be coupled to a drive member **250** as in the embodiments previously described. In FIG. **25**, the forces applied to switch **292** by shuttle **223** have caused switch **292** to be sheared into two separate pieces. Movable portion **293** of switch **292** is forced into the raised position where it is no longer aligned with locking member **270**. This allows for spring **278** to force locking member **270** to swing into channel **219**. In this position, drive member **250** is obstructed by locking member **270** should a user attempt to again actuate the surgical instrument. Stationary portion **294** of switch **292** remains in a substantially similar position.

(75) In embodiments, stapler cartridge **222** may include an additional switch **298** positioned on a proximal tail portion **224** of stapler cartridge **222** as best seen in FIG. **26**. In embodiments, the engagement of shuttle **223** with switch **298** upon distal translation of drive member **250** may serve as the reload detection point. In some instances, this configuration is desirable as the reload detection point is at a more proximal position than if reload detection was accomplished using switch **292**. In embodiments, switch **298** operates independently of locking member **270**, and switch **292** independently activates or disables locking member **270**.

(76) FIG. **27** illustrates, as an example, a top view of an operating room employing a robotic surgical system. The robotic surgical system in this case is a robotic surgical system **300** including a Console (“C”) utilized by a Surgeon (“S”) while performing a minimally invasive diagnostic or surgical procedure, usually with assistance from one or more Assistants (“A”), on a Patient (“P”) who is lying down on an Operating table (“O”).

(77) The Console includes a monitor **304** for displaying an image of a surgical site to the Surgeon, left and right manipulatable control devices **308** and **309**, a foot pedal **305**, and a processor **302**. The control devices **308** and **309** may include any one or more of a variety of input devices such as joysticks, gloves, trigger-guns, hand-operated controllers, or the like. The processor **302** may be a dedicated computer that may be integrated into the Console or positioned next to it.

(78) The Surgeon performs a minimally invasive surgical procedure by manipulating the control devices **308** and **309** (also referred to herein as “master manipulators”) so that the processor **302** causes their respectively associated robotic arm assemblies, **328** and **329**, (also referred to herein as “slave manipulators”) to manipulate their respective removably coupled surgical instruments **338** and **339** (also referred to herein as “tools”) accordingly, while the Surgeon views the surgical site in 3-D on the Console monitor **304** as it is captured by a stereoscopic endoscope **340**.

(79) Each of the tools **338** and **339**, as well as the endoscope **340**, may be inserted through a cannula or other tool guide (not shown) into the Patient so as to extend down to the surgical site through a corresponding minimally invasive incision such as incision **366**. Each of the robotic arms is conventionally formed of links, such as link **362**, which are coupled together and manipulated through motor controlled or active joints, such as joint **363**.

(80) The number of surgical tools used at one time and consequently, the number of robotic arms being used in the system **300** will generally depend on the diagnostic or surgical procedure and the space constraints within the operating room, among other factors. If it is necessary to change one or more of the tools being used during a procedure, the Assistant may remove the tool no longer being used from its robotic arm, and replace it with another tool **331** from a Tray (“T”) in the operating room.

(81) The monitor **304** may be positioned near the Surgeon's hands so that it will display a projected image that is oriented so that the Surgeon feels that he or she is actually looking directly down onto the operating site. To that end, images of the tools **338** and **339** may appear to be located substantially where the Surgeon's hands are located.

(82) The processor **302** performs various functions in the system **300**. One function that it performs

is to translate and transfer the mechanical motion of control devices **308** and **309** to their respective robotic arms **328** and **329** through control signals over bus **310** so that the Surgeon can effectively manipulate their respective tools **338** and **339**. Another important function is to implement various control system processes as described herein.

(83) Although described as a processor, it is to be appreciated that the processor **302** may be implemented in practice by any combination of hardware, software and firmware. Also, its functions as described herein may be performed by one unit, or divided up among different components, each of which may be implemented in turn by any combination of hardware, software and firmware.

(84) For additional details on robotic surgical systems, see, e.g., commonly owned U.S. Pat. No. 6,493,608, U.S. Pat. No. 6,671, and International Application WO 2017/132611. Each of these disclosures is herein incorporated in their entireties by this reference.

(85) FIG. **28** illustrates, as an example, a side view of a simplified (not necessarily in proportion or complete) illustrative robotic arm assembly **400** (which is representative of robotic arm assemblies **328** and **329**) holding a surgical instrument **450** (which is representative of tools **338** and **339**) for performing a surgical procedure. The surgical instrument **450** is removably held in tool holder **440**. The arm assembly **400** is mechanically supported by a base **401**, which may be part of a patient-side movable cart or affixed to the operating table or ceiling. It includes links **402** and **403** which are coupled together and to the base **401** through setup joints **404** and **405**.

(86) The setup joints **404** and **405** in this example are passive joints that allow manual positioning of the arm **400** when their brakes are released. For example, setup joint **404** allows link **402** to be manually rotated about axis **406**, and setup joint **405** allows link **403** to be manually rotated about axis **407**.

(87) Although only two links and two setup joints are shown in this example, more or less of each may be used as appropriate in this and other robotic arm assemblies in conjunction with the present invention. For example, although setup joints **404** and **405** are useful for horizontal positioning of the arm **400**, additional setup joints may be included and useful for limited vertical and angular positioning of the arm **400**. For major vertical positioning of the arm **400**, however, the arm **400** may also be slidably moved along the vertical axis of the base **401** and locked in position.

(88) The robotic arm assembly **400** also includes three active joints driven by motors. A yaw joint **410** allows arm section **430** to rotate around an axis **461**, and a pitch joint **420** allows arm section **430** to rotate about an axis perpendicular to that of axis **461** and orthogonal to the plane of the drawing. The arm section **430** is configured so that sections **431** and **432** are always parallel to each other as the pitch joint **420** is rotated by its motor. As a consequence, the instrument **450** may be controllably moved by driving the yaw and pitch motors so as to pivot about the pivot point **462**, which is generally located through manual positioning of the setup joints **404** and **405** so as to be at the point of incision into the patient. In addition, an insertion gear **445** may be coupled to a linear drive mechanism (not shown) to extend or retract the instrument **450** along its axis **463**.

(89) Although each of the yaw, pitch and insertion joints or gears, **410**, **420** and **445**, is controlled by an individual joint or gear controller, the three controllers are controlled by a common master/slave control system so that the robotic arm assembly **400** (also referred to herein as a “slave manipulator”) may be controlled through user (e.g., surgeon) manipulation of its associated master manipulator.

(90) While several embodiments have been shown in the drawings, it is not intended that the disclosure be limited thereto, as it is intended that the disclosure be as broad in scope as the art will allow and that the specification be read likewise. Therefore, the above description should not be construed as limiting, but merely as exemplifications of presently disclosed embodiments. For example, the invention is not limited to the mechanisms described herein for identifying and/or deactivating stapler cartridges. Other suitable devices or mechanisms are described in co-pending and co-owned International Patent Application No. PCT/US2019/66530, filed Dec. 16, 2019 and

entitled “SURGICAL INSTRUMENTS HAVING MECHANISMS FOR IDENTIFYING AND/OR DEACTIVATING STAPLER CARTRIDGES”, the entire disclosure of which is incorporated herein by reference in its entirety.

(91) Thus, the scope of the embodiments should be determined by the appended claims and their legal equivalents, rather than by the examples given.

(92) Persons skilled in the art will understand that the devices and methods specifically described herein and illustrated in the accompanying drawings are non-limiting exemplary embodiments. The features illustrated or described in connection with one exemplary embodiment may be combined with the features of other embodiments. Various alternatives and modifications can be devised by those skilled in the art without departing from the disclosure. Accordingly, the present disclosure is intended to embrace all such alternatives, modifications and variances. As well, one skilled in the art will appreciate further features and advantages of the present disclosure based on the above-described embodiments. Accordingly, the present disclosure is not to be limited by what has been particularly shown and described, except as indicated by the appended claims.

Claims

1. A surgical system comprising: a stapling instrument comprising a shaft, an end effector with first and second jaws, a switch disposed on, or within, the end effector, and a drive member configured to translate distally through the shaft, wherein the drive member is configured to contact the switch such that the switch provides a detectable resistance upon engagement with the drive member, wherein the switch is movable-relative to the first and second jaws by engagement with the drive member; and a controller operatively coupled to the surgical stapling instrument, the controller being configured to read the detectable resistance.
2. The system of claim 1, further comprising a staple cartridge having one or more staples removably loadable into one of the first and second jaws of the end effector, wherein the controller identifies the staple cartridge based on the detectable resistance.
3. The system of claim 2, wherein the drive member contacts the switch at an axial position of the drive member, and wherein the controller identifies the staple cartridge based on said axial position.
4. The system of claim 2, wherein the drive member comprises one or more inclined distal ramps and the switch has a contact portion configured to contact the one or more distal ramps upon distal translation of the drive member through the end effector.
5. The system of claim 4, wherein the contact portion of the switch is disposed at a height relative to the drive member such that the inclined distal ramp of the drive member is located at an axial position upon contact with the contact portion.
6. The system of claim 2, wherein the switch is disposed on the staple cartridge.
7. The system of claim 1, wherein the controller is part of a robotic surgical system operatively coupled to the stapling instrument.
8. The system of claim 1, further including a locking member movable from a disabled position permitting distal translation of the drive member through a staple firing stroke, to a locking position inhibiting distal translation of the drive member through the staple firing stroke.
9. The system of claim 1, further comprising a first staple cartridge comprising a first switch and a second staple cartridge comprising a second switch, wherein the drive member is configured to contact the first switch and the control unit is configured to identify the first staple cartridge and to contact the second switch and the control unit is configured to identify the second staple cartridge.
10. A control system for a surgical stapling instrument, the system comprising: a first controller coupled to an actuator for translating a drive member through a channel in an end effector of the surgical stapling instrument such that the drive member contacts and moves a switch from a first position, wherein the switch at least partially extends into the channel, to a second position,

wherein the switch is lateral of the drive member; and a second controller operatively coupled to the drive member and configured to detect a resistance encountered by the drive member as the drive member contacts the switch.

11. The control system of claim 10, wherein the second controller is configured to identify a staple cartridge within the instrument based on the resistance encountered by the drive member.
 12. The control system of claim 11, wherein the second controller is configured to identify the staple cartridge based on an axial position of the drive member when the drive member encounters the resistance.
 13. The control system of claim 10, wherein the drive member comprises one or more inclined distal ramps and the switch has a contact portion configured to contact the one or more distal ramps upon distal translation of the drive member through the end effector.
 14. The control system of claim 13, wherein the drive member contacts the switch at an axial position of the drive member, and wherein the second controller identifies the staple cartridge based on said axial position.
 15. The control system of claim 10, wherein the first and second controller are remote from the surgical instrument and form part of a robotic control system.
 16. A method for controlling a surgical instrument comprising: translating a drive member through a shaft of the instrument and an internal channel of an end effector of the instrument such that the drive member contacts and moves a switch extending into the internal channel; detecting a resistance encountered by the drive member upon contact with the switch; and identifying a removable unit in an end effector of the instrument based on said resistance.
 17. The method of claim 16, further comprising loading a staple cartridge into the end effector of the instrument and identifying the staple cartridge based on said resistance.
 18. The method of claim 17, wherein the drive member comprises one or more inclined distal ramps and the switch has a contact portion configured to contact the one or more distal ramps upon distal translation of the drive member through the end effector.
 19. The method of claim 18, further comprising identifying the staple cartridge based on an axial position of the drive member when the drive member encounters resistance with the contact portion of the switch.
 20. The method of claim 16, wherein the resistance comprises a torque spike.
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